

Towards the Safety of Film Viewers from Sensitive Content: Advancing Traditional Content Warnings on Film Streaming Services

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ABSTRACT

Traditional content warnings on film streaming services are limited to warnings in the form of text or pictograms that only offer broad categorizations at the start for a few seconds. This method does not provide details on the timing and intensity of sensitive scenes. To explore the potential for improving content warnings, we investigated users' perceptions of the current system and their expectations for a new content warning system. This was achieved through participatory design workshops involving 11 participants. We found users' expectations in three aspects: 1) develop a more nuanced understanding of their personal sensitivities beyond content sensitivities, 2) enable a trigger-centric film exploration process, and 3) allow for predictions regarding the timing of scenes and mitigating the intensity of sensitive content. Our study initiates a preliminary exploration of advanced content warnings, incorporating users' specific expectations and creative ideas, with the goal of fostering safer viewing experiences.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Participatory design; User interface design.**

KEYWORDS

Film Viewing, Content Warning, Triggering Content, User Experience, Streaming Services, Participatory Workshop, Interface Design

ACM Reference Format:

Soyeong Min, Minha Lee, and Sangsu Lee. 2024. Towards the Safety of Film Viewers from Sensitive Content: Advancing Traditional Content Warnings on Film Streaming Services. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3650980>

1 INTRODUCTION

Film streaming services allow users to watch films anywhere, anytime. Currently, many countries utilize content rating systems to signal the presence of sensitive content in films, such as violence, sex, and drugs [2, 7, 14–16]. However, triggering content, not included

in these ratings, can evoke past traumas or negative emotions in viewers, such as eating disorders or flashing lights, causing mental distress. Therefore, previous studies have highlighted the necessity of informing viewers about such content [6, 11, 20]. Many viewers seek to pinpoint triggering scenes in specific films through online review sites and communities, such as IMDb and DoesTheDogDie [20] prior to watching. It indicates the need for film streaming services to provide this information in advance. However, most film streaming services currently offer only pictograms that briefly indicate the reasons behind a film's rating at the start, disappearing after a few seconds. These services often do not cover all topics requiring content warnings, such as stigma, disturbing content, death, crime, abuse, or sociopolitical issues [6], or they show warning descriptions only for some films (refer to Figure 1). A few researchers have explored more effective ways of delivering content warning messages for certain triggers in films. For instance, South et al. have suggested improvements in flashlight warning messages based on a thematic analysis of users' reviews on an online community [20].

Text-based content warnings remain controversial regarding their effectiveness in reducing negative emotions and advising decisions to avoid viewings [4, 5, 13], thereby, recent scholars examining traumatic content in social media communities [10, 11, 17, 19] have suggested trauma-informed design implications for better user experience. These include notifying users who post triggering content about potential perceptions of their posts by others or automatically filtering words. However, applying these implications to film viewing is challenging due to the unique nature of films. A film, lasting from one to three hours, may include various sensitive scenes and is particularly vulnerable to spoilers, confining descriptive explanation. Moreover, streaming services provide content one-sidedly, making it hard for users to respond to sensitive content unlike in social communities, where users can freely add warnings to posts and replies. Existing efforts to create a safer ecosystem from sensitive content have been active in social communities from the perspective of designers and researchers, yet few have focused on film viewing with including users' voices in creating a trauma-informed approach [19], showing a dearth in research.

This study aims to explore new warning systems through user expectations. A participatory design workshop was conducted to gather user perceptions of current content warnings and to investigate their expectations for future warning methods. We recruited participants who have previously experienced mental stress due to sensitive content and also have background in user interface design from the intentions to encourage active ideation and fluent visual articulation, while also anticipating a diversity of unique ideas not currently present. After conducting two workshops with

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CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
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ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650980>

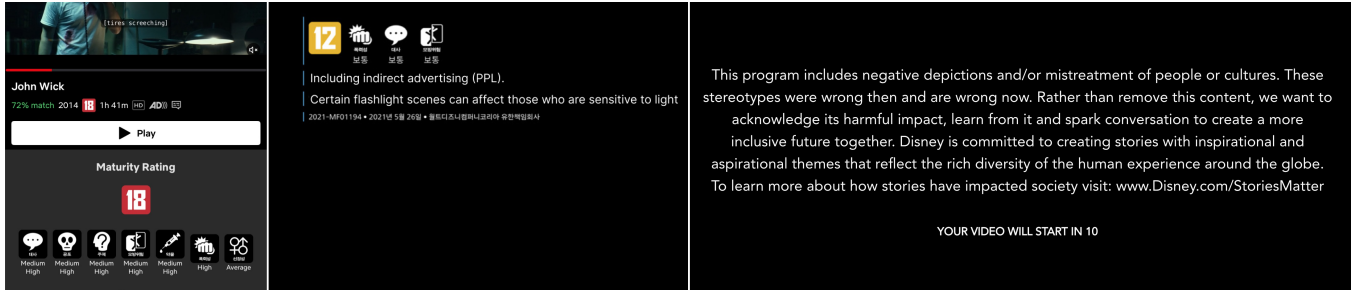


Figure 1: Content warnings in film streaming services (Left to right): Pictograms to show content label criteria from Netflix, Content warnings for flashlight from Disney+, and Content warnings only in [Dumbo] and [Peter Pan] from Disney+ [9].

11 participants, we discovered that they did not find current content warnings helpful and envisioned a comprehensive warning system in three main aspects: 1) understand their content sensitivity and preferred handling methods, 2) facilitate a trigger-centric film exploration process, and 3) enable them to anticipate scene timings and comfortably view scenes with low intensity. Our contributions include uncovering under-explored user expectations for a future content warning system and introducing innovative ideas. These insights represent a valuable effort towards creating an environment where more users can enjoy safer film-viewing experiences.

2 RELATED WORKS

The HCI community has documented the effects of sensitive content on individuals and explored measures to mitigate trauma-related distress in online ecosystems [17–19, 21]. Content and trigger warnings are utilized across multiple sectors, including film [3, 6, 21]. With the increasing use of content warnings, their formats are evolving across various media. Gupta observed varied content warnings on online platforms; many image-based platforms, such as Instagram and Facebook, blur sensitive images and provide an option to reveal them with warning messages [11]. On the other hand, video platforms like YouTube and TikTok introduce a pop-up window asking users' consent before displaying sensitive content. Numerous film streaming services briefly display warning messages at the start of a film (refer to Figure 1). The messages are varied in four categories: 1) specifying types of content, 2) alerting to potential emotional reactions, 3) indicating specific groups that should be careful, and 4) suggesting the best actions to take [3]. However, Bridgland et al. revealed no significant effect on reducing mental shock or effecting viewers' decisions to avoid sensitive content when a content warning was provided in varied message forms [3].

Efforts have been made to foster safe user experiences in considering triggering contents in a design perspective, such as exploring user interface design. Randazzo & Ammari have explored social media features that assist trauma survivors who struggle to articulate their experiences, introducing concepts like anti-personalization and directed word filters to better meet their needs [17]. Scott et al. have emphasized the risk of experiencing trauma within social communities, advocating for trauma-informed approaches for designers, moderators, and companies, including solutions like nudges

for users to reconsider traumatic content [19]. Melissa provided insights into designing for individuals who have experienced trauma [10]. While discussions on creating well-being from sensitive content have primarily occurred in social media interactions, a notable research gap exists in developing new design approaches that extend beyond current content warnings for film audiences. This gap has led our study to investigate the demand for innovative warning systems in film streaming services.

3 METHODOLOGY

The aim of this study was to understand participants' perceptions of the existing content warning method and their expectations for improvements in the future warning system. A participatory design workshop was conducted with groups to actively discuss their experiences and explore ideas, taking into account the perspectives of various viewers.

3.1 Participants

We looked for participants who required content warnings and were capable of envisioning their expectations for an improved film viewing environment on streaming services. Therefore, participants needed to meet two criteria: 1) adults over 19 who have experienced mental stress rendering them unable to watch specific scenes, and 2) those with experience in user interface design. A total of 11 participants were recruited, with an average age of 25.2 (age range: 22–29) (refer to Table 1). The types of sensitive content reported by participants were categorized as follows, aligning with a systematic typology of topics requiring content warnings identified in previous research [6]: 1) violence (e.g., blood tests, weapons, and bodily mutilation), 2) nervousness (e.g., jumpscare and tension-inducing sounds), 3) objectionable objects, 4) abuse, 5) disasters, 6) sociopolitical issues (e.g., war and Nazism), 7) flashing lights.

3.2 Procedure

This experiment was conducted face-to-face and comprised of four sessions: 1) group discussion, 2) brainstorming, 3) idea concretization, and 4) idea sharing. From the first three sessions, participants shared their film viewing experiences and visualized ideas for a new content warning system reflecting their expectations within their assigned groups. The final session facilitated idea exchange

Table 1: Participants' Information via Screening Survey Responses

Group	No	Age	Sex	Design Experience	Content Sensitivity
1	P1	24	M	Graduate student in industrial design major	Gross objects (caterpillars), war
	P2	22	F	Graduate student in industrial design major	Tension-inducing sounds or knife sound, jumpscare, animal abuse, child abuse, bodily harm
	P3	24	F	Currently working in the UX/UI design field	Brainwashing and abuse in religion, transportation accidents
2	P4	26	F	Graduate student in industrial design major	Child abuse, historical events(like Nazism), films inducing extreme sad feelings, gross objects(caterpillars), dirt
	P5	26	F	Graduate student in industrial design major	Jumpscare, flash lights, child abuse
	P6	24	M	Have conducted at least one UX/UI design project before	Animal abuse, blood, gross objects
3	P7	26	M	Graduate student in industrial design major	Blood, bodily harm
	P8	25	F	Have conducted at least one UX/UI design project before	Bodily harm, human rights violence, sexual abuse
	P9	29	F	Currently working in the UX/UI design field	Jumpscare, dirt, detailed bodily harm, earthquake
4	P10	26	M	Have conducted at least one UX/UI design project before	Jumpscare, detailed bodily harm
	P11	25	F	Have conducted at least one UX/UI design project before	Knife sound, bodily harm with knives, detailed death scene

between two groups, which will be explained more in detail in the following. To promote active discussion and ideation, researchers grouped two or three participants with differing sensitivity classifications based on survey responses (refer to Table 1). Four groups were formed and scheduled into two studies (Group 1 & 2, Group 3 & 4). At the start of each session, the moderator outlined the task ahead and provided necessary materials and templates. The initial session, group discussion, aimed to gather participants' perceptions of current content warnings, utilizing classified topics [6] and four questionnaires addressing difficulties caused by specific scenes, resultant actions taken, and the limitations and impacts of content warnings. This set the stage for a 20-minute open discussion. The second session, brainstorming, engaged participants in generating ideas for a new warning system for 20 minutes. They used sticky notes within a digital collaboration tool, Figjam. This tool enabled participants to view each other's contributions, freely arrange notes, and vote anonymously. In the third session, the groups were assigned to further develop one or two of their ideas, making them more concrete and tangible by utilizing either paper-based or digital methods, for approximately 50 minutes. Lastly, the groups presented their ideas and exchanged feedback on the advantages and disadvantages, encouraging a critical reassessment of their concepts. Each experiment lasted roughly 1 hour and 50 minutes, with all procedures audio-recorded and photographed, with participants' consent.

3.3 Data Collection & Analysis

The audio recordings were transcribed, and the outcomes of the tasks were transformed into textual data for analysis. Additionally, the visualized ideas were converted into text descriptions. For thematic analysis, we utilized the open coding method [8]. This analysis concentrated on the understanding general perceptions of content warnings and anticipation for enhancements in their delivery. The initial codes were generated by the first author, and the analysis were refined through multiple iterations with two other researchers, employing the affinity diagram technique [12].

4 FINDINGS & DISCUSSIONS

In this section, we present our major findings with discussions. Initially, we briefly examine the perceptions of current content warnings. Following that, rather than proposing specific design implications, we aim to convey the needs and expectations of the participants for a new warning system, drawing on their concrete ideas generated during the workshops.

4.1 User Perceptions of Current Content Warnings

We found that, although necessary, the current content warning is ineffective, a finding that is consistent with previous studies [3, 20]. Nine out of eleven participants emphasized their reluctance to forego watching films even if they contain triggering scenes, with one participant stating, *"I feels it is too valuable to give up on the entire movie just because of that one scene"* (P7). In this context, all participants reported experiences that had a traumatic impact or caused mental distress.

"There was a scene where worms came out of his head or hands. It was extremely traumatic for me. Anything related to worms has become a symbol of horror for me." (P1)

While previous studies only focused on individuals who struggle with watching sensitive content [10, 11, 17, 19, 20], P3, P10, and P11 noted that content warnings could serve not only to create a safe viewing environment for sensitive viewers but also to assist those with higher tolerance in effortlessly finding intense films they enjoy.

"If films are organized by keywords, people who are vulnerable at that point might choose not to watch, and those who like it might easily find such content because of it." (P3)

However, current content warnings failed to meet the participants' expectations due to their lack of descriptive effectiveness and visibility as reported in [20]. P2, P3, P5, and P11 found it challenging

to predict the intensity of the scenes based solely on warning messages. P1, P3, P4, P5, and P6 noted that the short display of warnings led to reduced awareness, mentioning that warnings displayed at the beginning of the film made it difficult to opt out of watching. Additionally, there was disappointment that specific sensitivities, such as ‘a scene where a main character is dizzy’ (P4) and ‘loud noise’ (P7), were not addressed in the warnings. Despite attempts to categorize topics used in content warnings including these issues [6], it was evident that the topics are currently not well-reflected. We highlighted the need to explore the next steps to convey content warnings beyond current methods.

"It's just a warning pop-up saying, 'There are provocative scenes, so if you are uncomfortable with it, please skip it.' [...] But when I actually watched it, sometimes it's as expected, but often it's more shocking than I anticipated." (P11)

4.2 Users Expect Streaming Services to Better Understand Their Content Sensitivity and Preferred Handling Methods

Participants expressed a desire for streaming services to adopt a personalized approach to managing sensitive content, aligning with the previous discussions [6, 11]. Given the variation in content sensitivities and preferred handling methods among participants, a detailed understanding of these differences appears necessary. We identified five features, reflecting the diverse traits and needs of participants, marking an initial step towards developing tailored content warnings for viewers.

4.2.1 Trigger Scope. Participants desired the warning system to precisely understand not only which content they are sensitive to but also the extent of their sensitivity to each type of content, naming this feature *trigger scope*. Highlighting their unique sensitivities, similar to allergies as mentioned by P7, P8, and P9 (Group 3) (refer to Table 1), they stressed the importance of recognizing individual tolerance levels on a continuous scale, beyond a simple binary of ‘sensitive’ or ‘not.’ Eight out of eleven participants anticipated a warning system that features a concise and engaging process for gathering information. They showed their expectations with two solutions: proactive input by the users and automatic refinement by the system. P10 and P11 (Group 4) looked forward to conveying their *trigger scopes* to the system in an enjoyable way, using the concept of avatars shown in Figure 2. They could place metaphorical stickers, such as knives and guns, on their avatars to indicate their sensitivities. P7, P8, and P9 (Group 3) anticipated that the system would autonomously improve its understanding of them. For example, if they stopped a film midway or down-voted it, the system would automatically adjust their *trigger scopes* by analyzing similar scenes from those films.

4.2.2 Sensory Sensitivity. Each participant exhibited varying sensitivities to the visual and auditory elements of scenes, seeking assistance that considers these sensitivities during viewing. P7 mentioned that looking away from sensitive scenes was usually sufficient but highlighted a particular vulnerability to loud noises. Conversely, P8 and P11 observed that merely hearing violent scenes

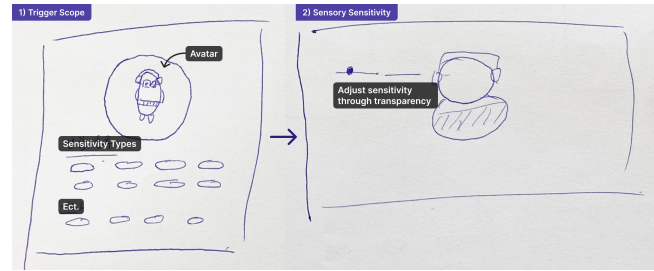


Figure 2: Group 4’s Idea: Expressing their content sensitivities by customizing an avatar.

could be mentally distressing and induce fear. In response, P7 expected that the sound could be automatically lowered in scenes with sudden loud noises, while P10 suggested expressing their sensory sensitivity level by adjusting the transparency of glasses and earplugs, representing their sensory sensitivities, in the character customization process described in Section 4.2.1.

4.2.3 Interaction Autonomy. While watching films, participants expressed two distinct preferences for interacting with the interfaces. P9 and P10 wanted the autonomy to manually adjust scenes in terms of visual and audio aspects. P9 emphasized that with a sufficient warning before watching, it would be better for viewers to have freedom to control screens, like skipping scenes. In contrast, the other participants preferred automated adjustments by the system, allowing them to focus on watching without direct interaction. P1 voiced concerns that having to make quick decisions could feel pressuring during viewing. Accordingly, eight participants expected that the warning system would automatically adjust scenes as described in Section 4.4.

4.2.4 Message Specificity. Preferences for the specificity of warning messages varied among participants. P4 and 7 preferred minimal information to avoid imagining scenes, so they even avoided plot summaries and trailers. In contrast, the others wanted details about sensitivities, such as the level of violence. P8, P9, P10, and P11 sought as specific information as possible about sensitive content, with P9 stating, “I would like to know even if it is a spoiler when a scene I dislike is about to appear” (P9). For those particularly sensitive to certain scenes, the first group preferred warnings with just keywords rather than detailed plot descriptions, while the second group hoped for highly described messages.

4.2.5 Desirable Timing for Warnings. There were varying preferences regarding when participants wanted to receive warnings. P10 wanted to receive warnings before watching films and not to receive additional information during the viewing. In contrast, other participants (P1, P2, P5, P7, and P8) expected to receive warnings while viewing. This finding highlights the potential for varied timing in warning delivery, contrasting with previous research that focused solely on pre-viewing warnings [3–5, 11, 13, 20].

4.3 Users Expect Streaming Services to Enable a Trigger-Centric Film Exploration Process

Participants expected an environment that allowed them to precisely check sensitive content through personalized information and to search for films with consideration of that content. These expectations emerged from two specific approaches.

4.3.1 Content Warning Without Describing Content. While traditional content warnings were based solely on the content of the corresponding film itself [3, 11, 20], participants had two expectations for a new content warning without describing scenes. Firstly, P1 and P5 expected to use their viewing histories and comparing them to the target film. The second approach of P1, P9, and P10 was analyzing other users' reviews and mapping expressions associated with triggering keywords. The participants anticipated that these approaches could help them determine the intensity of the scenes better without spoilers.

"People can see how intense the contents are by comparing them to movies they've already seen. For instance, it has a similar level of violence to 'Saw.'" (P5)

"When collecting words frequently used alongside specific content keywords in comments or reviews, users can see which expressions are close to the keywords [...] they can decide whether they can watch it. They can find out it will be extremely intense if the keyword is mapping these expressions." (P1)

4.3.2 Sensitive Content-Centered Film Searching Experience. Participants revealed expectations for a new film exploration experience through two specific ideas. Firstly, based on answers from P1 and P4, integrating their *trigger scope* into current recommendation algorithms could allow them to find preferred films while avoiding certain triggers. However, P9 expressed concerns about the automatic exclusion of films with sensitive content without her consent, saying "What if all the movies are just removed with those scenes from my home feed? [...] I don't want that either" (P9). This consideration is crucial for further developing this expectation. Secondly, P4, P5, and P6 (Group 2) envisioned an interactive search process beyond traditional title or genre-based methods. The idea of removing trigger keywords like removing items from a shopping cart (refer to Figure 3-a) would adjust the top-recommended film to one without the excluded triggers.

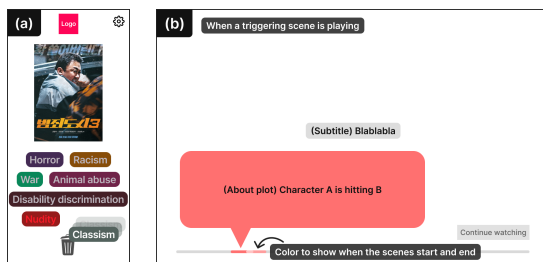


Figure 3: Participants' ideas: (a) Finding films by removing keywords from group 2, (b) Replacing scenes with text with colorizing playback bar from group 1.

4.4 Users Expect to Anticipate Scene Timings and Comfortably View Scenes with Low Intensity

Previously, streaming services commonly did not intervene while streaming. However, our finding showed that all participants except P10 wanted safe viewing, even with active intervention from the warning system. They had two expectations: precisely predicting the timings of sensitive scenes and comfortably viewing through less intense scenes.

4.4.1 Predicting Precise Timings of Scenes. The need to know the exact timing and intensity of scenes was captured by the idea of P1, P2, and P3 (Group 1), which is colorizing segments of the playback bar red and varying saturation in Figure 3-b. P3 believed this would help them respond proactively to the scenes.

"When the intensity is expressed on the playback bar [...], users can just look at it and skip it at their discretion, so I think this method is also good." (P3)

4.4.2 Understanding Scenes with Text. Participants had taken steps to avoid sensitive content, such as reducing the screen size or skipping scenes that they found inconvenient. For instance, P9 often repeatedly skipped scenes without knowing the end of the scene, leading to a lack of understanding of the plot. As mentioned in Section 4.2.3, eight of eleven participants expected the warning system to automatically identify segments needing adjustment and to take actions like blurring, lowering resolution, or skipping the scene. They also wanted specific textual summaries of these scenes and the option to watch the original scene if chosen, aiming to lessen inconvenience and maintain a level of viewer autonomy (refer to Figure 3-b).

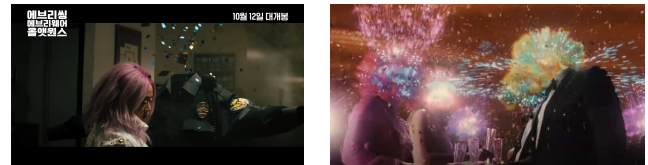


Figure 4: Examples of replacing an explosion with fireworks or sparkling (Left to right): [Everything Everywhere All at Once] [22] and [Kingsman] [1].

4.4.3 Watching Comfortably with Less Intense Scenes. P7, P8, P9, P10, and P11 (Group 3 & 4) desired a more comfortable viewing experience, envisioning the idea of using generative AI to replace original intense scenes with milder ones. These users referred to examples from two films where fireworks and sparkles substituted for bloody explosions (refer to Figure 4). We identified the importance of AI technology in playing a crucial role in transforming these expectations into implications in further work.

5 LIMITATIONS & FUTURE WORKS

This study explored the potential of a new content warning system for streaming services but did not validate its efficacy and practical feasibility. Consequently, future work is expected to refine

and validate the effects of the designs through further prototyping and testing. Additionally, while seeking to extract overall expectations, this study did not fully address the concerns of users who are extremely sensitive to triggering content. Given the importance of content warnings for these users, future research should delve more deeply into their needs. Furthermore, based on our findings, we recognized the differences between stakeholders. For instance, as discussed in Section 4.4, identifying potentially sensitive scenes in films for screening them might cause mental distress to moderators [19], and producers may be resistant to such measures. This indicates that complex considerations may be required for practical applications. Lastly, further exploration of how the new content warning system can impact user experience, including potential disruptions, will be necessary.

6 CONCLUSION

This study delved into user needs and expectations for a novel warning system, building upon their perceptions of existing content warnings in film viewing experiences. Through the workshops, we identified previously unnoticed user needs and specific expectations grounded in tangible and visually-oriented interface ideas. This study revealed users expected the system to: 1) understand their content sensitivity and preferred handling methods, 2) enable a trigger-centric film exploration process, and 3) enable them to anticipate scene timings and comfortably view scenes with low intensity. Despite several limitations noted earlier, this research contributes to the HCI community by initiating a preliminary investigation into users' detailed expectations and innovative interface ideas. It lays the groundwork for developing a new warning system tailored to film viewing, aiming to create safer and more inclusive experiences.

ACKNOWLEDGMENTS

We would like to express our gratitude to the participants for their deep engagement during the workshop and their efforts to explore improved ideas for the warning-informed design approach presented in this study.

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A APPENDIX. PHOTOS OF WORKSHOP SITE

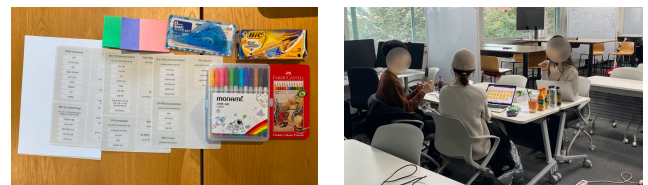


Figure 5: (Left to right) Typologies in group discussion and drawing tools in visualizing idea session, and a photograph when participants share their idea within group.